

Specification

Core

Function	Description
MCU	Module : ESP32-S3 N16R8 Processor : Xtensa® LX7 dual core 32-bit up to 240 MHz SRAM : 512KB ROM : 384KB Flash : 16MB PSRAM : 8MB RTC SRAM : 16KB
Wi-Fi	Specification : IEEE 802.11b/g/n Frequency : 2.4G Band width : 20MHz、 40MHz
Bluetooth	Specification : Bluetooth 5、 Bluetooth mesh Frequency : 125 Kbps、 500 Kbps、 1 Mbps、 2 Mbps

On-board component

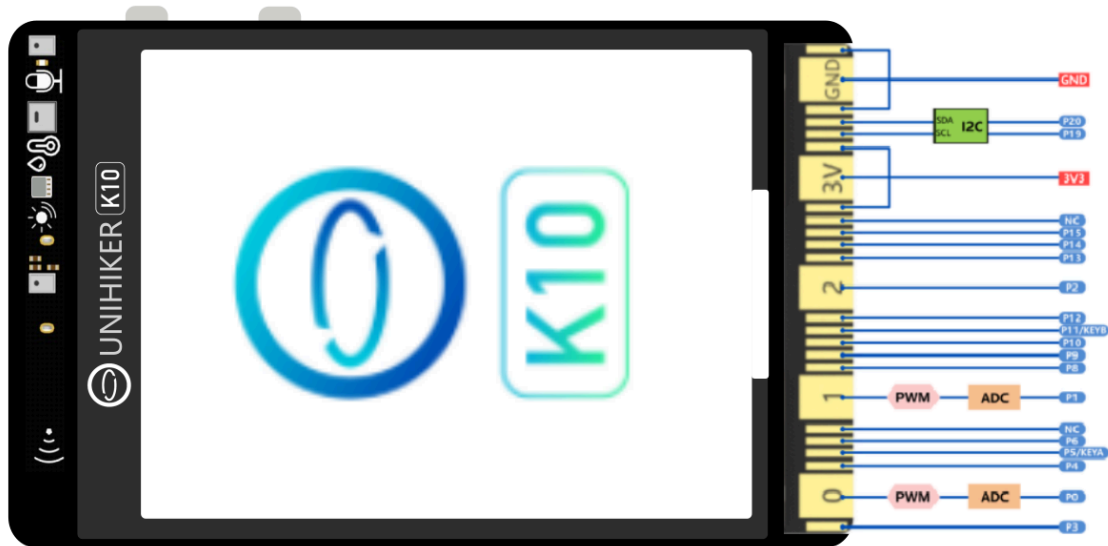
Function	Description
Temp&Humi Sensor(AHT20)	Temperature : -40-85°C Accuracy±0.3°C Humidity : 0-100%RH Accuracy±2%RH
Ambient Light Sensor(LTR303ALS)	Detection range : 0~64k Lux
Triaxial Acceleration Sensor(SC7A20H)	±2G/±4G/±8G/±16G

Function	Description
Display	2.8inch 240x320 Colored LCD ILI9341
Camera	2MP, 80°FOV GC2145
A, B, RST,BOOT button	A, B, RST,BOOT button
MEMS MIC	MEMS MIC x 2
Speaker	Power : 2W
WS2812 RGB LED	WS2812 RGB LED x 3

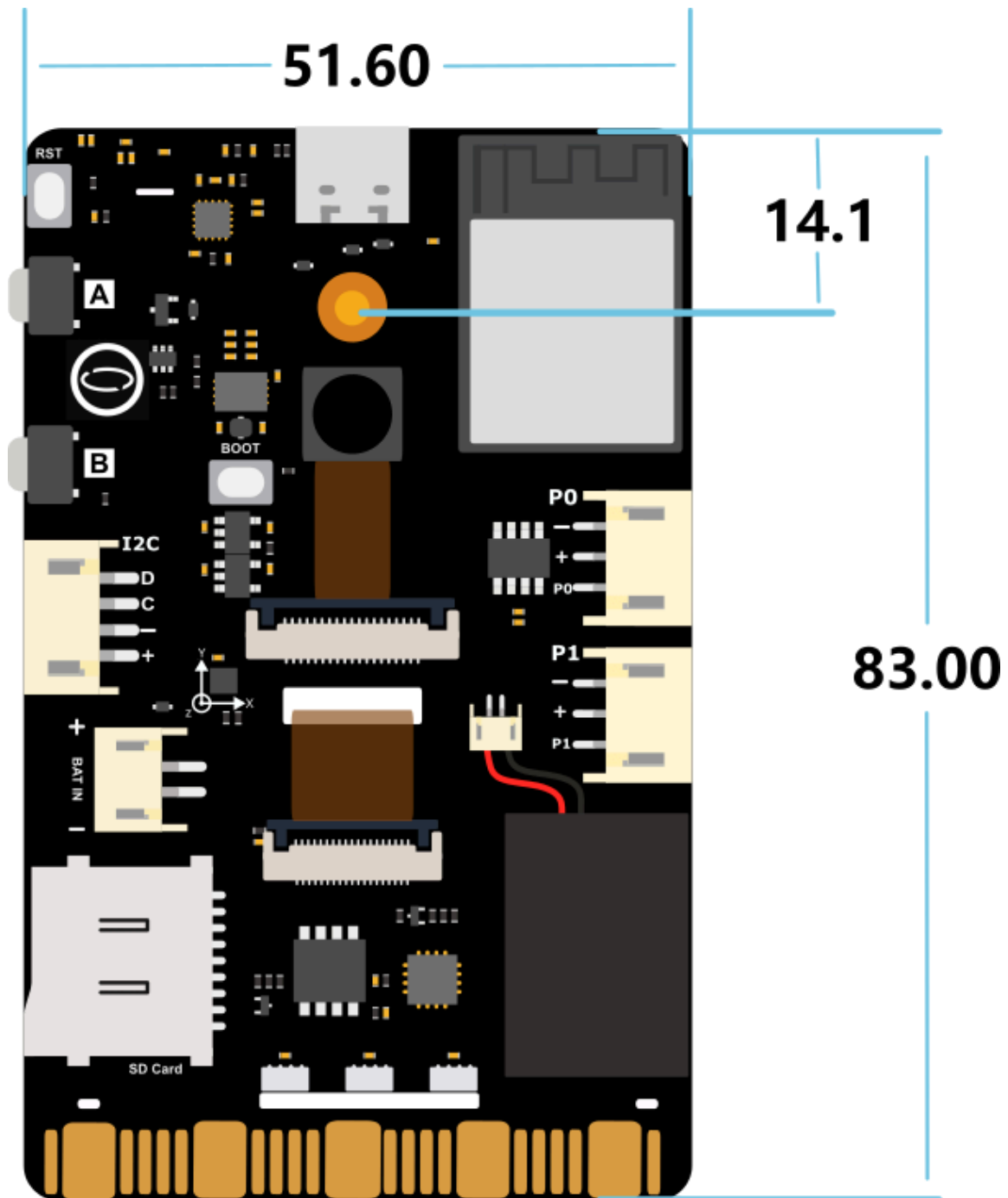
Interface

Function	Description
On-board interface	USB-C power(5V DC)/program upload interface 2Pin PH2.0 battery interface(3.0-6.0V DC) Recommended power source: 3.7V Lipo battery or three 1.5V AA/AAA cells 3Pin PH2.0 Full function GPIO x 2 4Pin PH2.0 I2C Self-ejecting TF card slot
EdgeConnector	Full function GPIO x 2 I2C x 1 Digital IO x 15

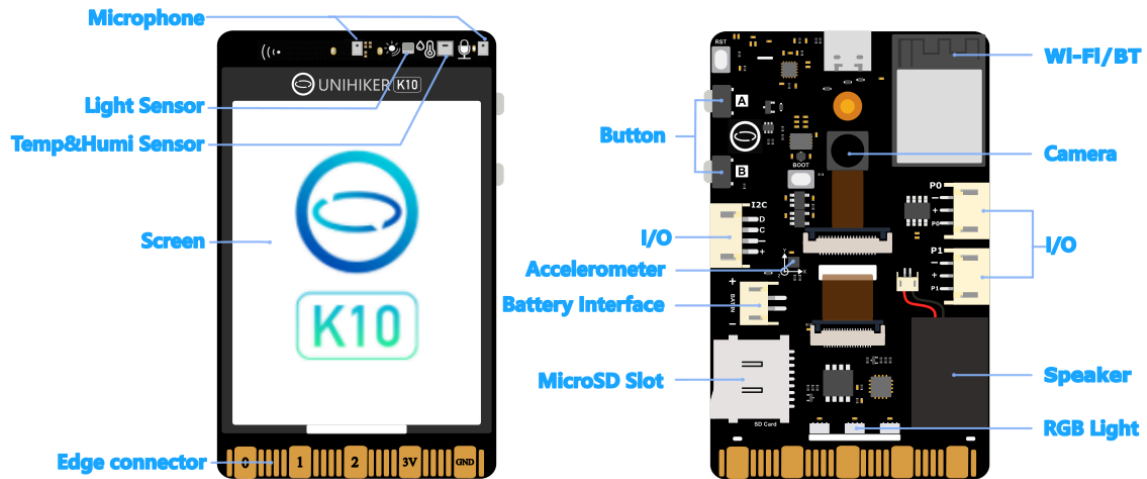
Boardover view



Dimensional Drawing



On-board resorce



System Framework

